

Urban Heat in Six Districts: Summer 2026 Field Study

Mara Lindqvist and Daniel Achterberg · City Climate Unit, Working Paper 14 (fictional)

Abstract. We measured surface and air temperatures in six districts of a coastal city from June to August 2026 and tested four cooling measures. Districts with little tree cover were the hottest by day and cooled least at night. Misting stations gave the largest local cooling but cost the most to run.

1 Introduction

Heat waves are now the deadliest weather hazard in the city. Older residents in flats without cross-ventilation are most at risk, and ambulance call-outs rose by a fifth on the hottest days of 2025.¹

The study asked two questions: which districts stay hottest, and which low-cost measures cool them most for each pound spent.

2 Methods

Air temperature was logged every ten minutes in each district, and surface temperature was measured with a handheld infrared camera at noon on twenty clear days.

District boundaries follow the census wards, merged where a ward had fewer than 4,000 residents.² Each measure was tested on at least two streets, with a matched street nearby as a control.

3 Results

Surface temperatures in the Eastgate district peaked at 47.2 °C on 19 July, the highest of the six districts measured. Tree cover in Eastgate is 6 per cent, against a city average of 19 per cent.

Night-time air temperatures stayed above 24 °C for nine nights in a row in the Old Market area, where most streets run east to west and trap the evening sun.

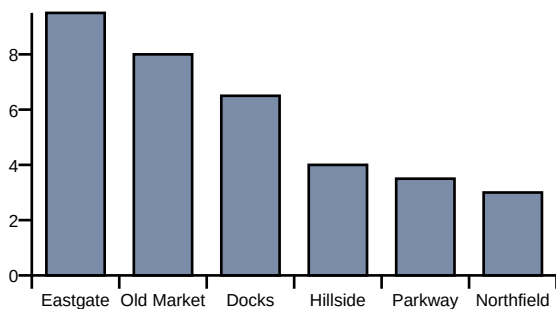


Figure 1. Hours above 30 °C on the hottest day, by district; Eastgate stayed above 30 °C for 9.5 hours.

The Hillside district, which has the most tree cover at 31 per cent, was on average 3.4 °C cooler at noon than Eastgate. The difference shrank to 1.1 °C after sunset.

Wind from the sea cooled the Docks district by about 2 °C on most afternoons, but only within 300 metres of the shore.

4 Cooling measures

Cool roofs, painted with a white reflective coating, were the cheapest measure per square metre. Street trees took two summers to give shade but cooled the pavement beneath them the most.

Misting stations cooled the air within five metres by up to 3.1 °C, but each used about 40 litres of water an hour.³ Residents asked for them at bus stops and outside the two clinics.

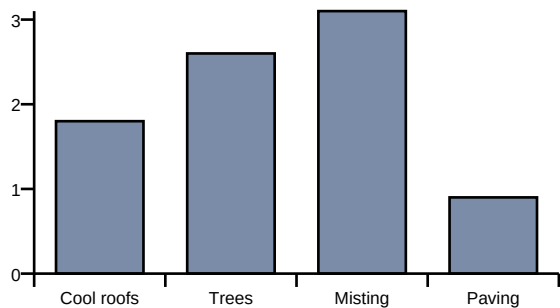


Figure 2. Cooling beside each measure at noon, in °C, averaged over 14 test streets; the paving was measured in the Parkway car park.

A survey of 1,140 residents found that 41 per cent of those in Eastgate changed their walking route on hot days, against 17 per cent citywide.

5 Discussion

The hottest districts are also the poorest: median household income in Eastgate is 38 per cent below the city's. Heat adds to health risks that are already higher there, and fewer homes have fans or blinds.

¹ Ambulance figures from the regional ambulance service's 2025 heat report.

² Wards were merged using the 2021 census boundaries.

³ Water use was read from each station's meter weekly.

Tree planting is slow to pay back but cheap once trees are established. In our model a street tree costs about £22 a year to look after from its fifth year, mostly for watering in dry summers.

Cool roofs suit the flat-roofed warehouses of the Docks district best. On pitched roofs the coating wore off within a single summer on two of the five test houses.

Misting is a stopgap. It helps people waiting at bus stops on the hottest afternoons, but it raises humidity, and the stations need daily checks for legionella.

6 Limitations

Twenty measurement days cover only part of the summer, and July 2026 was hotter than usual. Sensors in direct sun read up to 1.5 °C high until radiation shields were fitted on 3 July.

The survey over-represents residents who were at home in the daytime, who are also those most exposed to indoor heat. Answers were collected on paper and online, in English and Polish.

We did not measure indoor temperatures, which matter most for older residents at night; a follow-up study with 80 indoor loggers starts in June 2027.

7 Costs and recommendations

Table 1. Cooling measures tested in summer 2026, with the area treated, the cooling measured and the cost.⁴

Measure	Area treated	Cooling (°C)	Cost
Cool roofs	18,400 m ²	1.8	£24 per m ²
Street trees	2,150 trees	2.6	£310 per tree
Misting stations	12 sites	3.1	£9,800 per site
Permeable paving	6,700 m ²	0.9	£41 per m ²

We recommend planting street trees in Eastgate and Old Market first, and keeping misting stations for bus stops and clinics, where people wait outside.



Figure 3. The 12 misting stations installed in June 2026, clustered around bus stops on the two main roads into Eastgate.

The next study will repeat the measurements in summer 2027, after the first 600 trees are planted.

⁴ Costs are 2026 prices and exclude maintenance, which adds about 8 per cent a year.